Solar ESS Co., Ltd.
53 Moo 2, Nang Lae, Mueang, Chiang Rai 57100, Thailand
Tel. 063 151 6694, 089 632 5829 · LINE: solaress1
www.solaress.org · www.โซลาร์เอสส์.com

Brand Comparison — Inverter, Battery, Panel, Breakers & Cable

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Best picks at a glance

The best pick in each category for a 10 kW three-phase hybrid home or small-business system, chosen on specifications, certification, availability in Thailand and value.

Category	Best overall	Best in another area
Hybrid inverter 10 kW 3-phase	Solis S6-EH3P10K02-NV-YD-L	GoodWe GW10K-ET-L-G10: PV oversizing, backup power, warranty
Battery \~16 kWh	Solis IntelliHome-16kWh-OD	Dyness PowerBrick Plus: power and cycle life; Felicity: value per kWh
PV module	LONGi Hi-MO X10 Guardian Anti-Glare (single glass)	AIKO Comet 2U: 25-year product warranty, lightest
DC combiner set	Suntree	ABB: safety margin; FEEO: budget
AC combiner set	CHINT NXM-125S MCCBs + Suntree SPD + Suntree SQ8P ATS	Schneider EZC100H / ABB Formula: margin
Cables	Thai Yazaki THW (AC), LINK PV1-F (DC)	Lapp / Phelps Dodge: documented EN 50618
Mounting	Clenergy PV-ezRack	Unbranded kit: value; K2 Systems: quality
Conduit	Arrow IMC 1"	Panasonic: documented UL 1242
Grounding	Kumwell	Erico Cadweld: quality

Hybrid inverter: 10 kW, 3-phase

Pick: Solis S6-EH3P10K02-NV-YD-L. It is the best-value 48 V unit in Thailand, with 220 A battery current, IP66, 2× surge, unbalanced output on both ports, a generator port and dual backup ports. **Runner-up: GoodWe GW10K-ET-L-G10**, worth the extra cost where PV oversizing or a documented 10-year warranty matters. Sungrow and Huawei only take high-voltage batteries, so they don't fit a 48 V battery like the IntelliHome.

	Solis S6-EH3P10K02-NV-YD-L	GoodWe GW10K-ET-L-G10	Deye SUN-10K-SG05LP3	SRNE HESP48100SH3	Huawei SUN2000-10K-MAP0 (HV)
Battery	48 V, 220 A	48 V, 220 A	48 V, 210 A	48 V, 220 A	600–980 V, LUNA only
Max PV input	16 kW	20 kW	16 kW	15 kW	18 kW
MPPTs / current	2 / 20 + 40 A	3 / 20 A each	2 / 26 + 26 A	2 / 22 + 22 A	2 / 16 A
MPPT range / max DC	200–850 V / 1000 V	150–850 V / 1000 V	200–650 V / 800 V	200–650 V / 800 V	160–1000 V / 1100 V
Backup / surge	10 kW / 2× 10 s	11 kVA / 20 kVA 10 s	10 kW / 2× 10 s	10 kW / 20 kVA	10 kW, needs SmartGuard
Max efficiency	97.6 %	97.8 %	97.6 %	97.5 %	98.6 %
IP rating	IP66	IP66	IP65	IP65	IP66
Generator input	10 kW	20 kVA	Shared gen/smart-load port	Yes	No
Monitoring	SolisCloud	SEMS	DeyeCloud / SolarMan	Optional Wi-Fi	FusionSolar
Warranty (Thailand)	Confirm with distributor	10 yr	10 yr	5 yr	10 yr

Best in	Winner
PV oversizing	GoodWe, 20 kW on 3 MPPTs
Backup power	GoodWe, 11 kVA continuous, 20 kVA generator input
Efficiency	Huawei 98.6 % (HV only); GoodWe 97.8 % among 48 V units
Warranty	GoodWe, 10 years documented
Parallel / battery choice	Deye, up to 10 units in parallel, widest battery list

No public PEA approved-inverter list could be found online for these models; confirm with PEA or the distributor before ordering. The Sungrow SH10RT is high-voltage only and much more expensive, so it is left out.

Why Solis S6 stands out

The Solis S6 hybrids are built for heavy loads, oversized arrays and generator backup. Some of the strongest figures Solis publishes belong to its larger high-voltage models; the right-hand column gives what applies to the 10 kW low-voltage unit in this proposal.

Feature	What it means	10 kW LV model (S6-EH3P10K02-NV-YD-L)
High surge in backup mode	Starts pumps, air conditioners and motors without tripping	2× rated power for 10 s
High DC/AC ratio	More panels per inverter; surplus goes to the battery or loads	Up to 16 kW PV (160 %)
Generator port	Generator wired straight in, with automatic start/stop and peak shaving	10 kW generator input
Dual backup outputs	Critical and non-critical loads on separate ports, no extra load-shedding hardware	Yes
High-current inputs	Suits 600 W+ panels and large batteries	20 A / 40 A per MPPT; 220 A battery current
Fast UPS switchover	Computers and network gear keep running in a blackout	Under 10 ms, per Solis for S6 hybrids; confirm on the LV datasheet
Unbalanced output	Uneven loads across the three phases	Supported on grid and backup ports

Sources: Solis S6-EH3P(30-50)K-H-ND · Solis S6-EH3P(5-10)K-H · Solis launch notice · S6-EH3P(8-15)K02-NV-YD-L datasheet

Battery: about 16 kWh, 51.2 V LiFePO4

Pick: Solis IntelliHome-16kWh-OD. It is the only battery here built for the Solis S6 over CAN, so the BMS link and SolisCloud work without extra setup. It is also IP66 and IEC 62619 certified, uses 100 % of its capacity, and is good value per usable kWh. Its one weakness is about 8 kW continuous (157 A): it cannot carry the full 10 kW from battery alone, so a second unit is needed if night loads exceed 8 kW.

Model	Usable kWh (DoD)	Continuous charge / discharge	Cycles	Warranty	Rating	On the Solis S6-EH3P list
Solis IntelliHome-16kWh-OD	16 (100 %)	157 / 157 A, \~8 kW	>6,000 at 100 %	10 yr	IP66, outdoor, heater	Native Solis
LVTOPSUN LVTS-512314-G4	14.5 (90 %)	150 / 200 A	8,000 at 90 %	10 yr	IP65	No (only 100–300 Ah models)
Felicity FLA48314-Plus	15.2 (95 %)	160 / 200 A	≥8,000	7 yr in Thailand	IP65 (FLB)	No
Dyness PowerBrick Plus	15.3 (95 %)	200 / 200 A, \~10.2 kW	≥8,000 at 95 %	10 yr	IP65	Only "PowerBrick", not Plus

Model	Usable kWh (DoD)	Continuous charge / discharge	Cycles	Warranty	Rating	On the Solis S6-EH3P list
Deye SE-F16	14.4 (90 %)	157 / 230 A	6,000+	10 yr / 50 MWh	IP21, indoor only	No
Pylontech US5000 × 3	13.7 (95 %)	\~12 kW	>6,000	10 yr	IP20 rack	Yes, by name

Compatibility is checked against the Solis S6-EH3P(8-15)K02-NV-YD-L battery list (Aug 2025).

Best in	Winner
Solis compatibility and outdoor install	Solis IntelliHome (native, IP66)
Power	Dyness PowerBrick Plus, 200 A both ways (\~10.2 kW)
Cycle life	Dyness PowerBrick Plus, ≥8,000 cycles at 95 %
Avoid here	Deye SE-F16: indoor only and not on the Solis list

No model lists a cell grade or a TIS (มอก.) certificate. For any 314 Ah third-party battery, get written confirmation from Solis Thailand before ordering.

PV modules: LONGi vs AIKO

Pick: LONGi Hi-MO X10 Guardian Anti-Glare, single glass (LR7-72HVHF), 650 W. It is the only one of these panels with measured anti-glare data, ties AIKO on efficiency, and is the cheapest per watt in Thailand. Both brands use n-type back-contact cells with the same −0.26 %/°C heat loss and the same degradation (88.85 % at year 30), so heat and ageing are a tie.

Model	Cell	Efficiency	Voc / Isc	Max fuse	Weight	Glass	Product warranty
LONGi Hi-MO X10 Guardian LR7-72HVHF	HPBC 2.0	24.1 % (24.8 % at 670 W)	53.90 V / 15.29 A	25 A	28.5 kg	Single, anti-glare	15 yr
LONGi Hi-MO X10 Guardian LR7-72HVD	HPBC 2.0	24.06 %	54.22 V / 15.14 A	30 A	32.5 kg	Dual, anti-glare, bifacial	15 yr
LONGi Hi-MO X10 Scientist LR7-72HVH	HPBC 2.0	24.1 %	53.90 V / 15.29 A	25 A	28.5 kg	Single, standard	15 yr
AIKO Comet 2U G-MCH72Mw	ABC	24.4 % (24.8 % at 670 W)	54.40 V / 15.12 A	25 A	28.1 kg	Single, standard	15 yr, up to 25
AIKO Stellar 1N+ G-MCH72Dw	ABC	24.1 %	54.00 V / 15.06 A	30 A	32.3 kg	Dual, bifacial	not stated

Best in	Winner	Why
Anti-glare	LONGi Guardian	Reflectivity 0.9 % vs 2.9 %, EN 17037 "AAA" (LONGi); AIKO publishes no anti-glare model
Efficiency	Tie	Both 24.8 % at 670 W; AIKO slightly ahead at 650 W (24.4 % vs 24.1 %)
Product warranty	AIKO Comet 2U	Extendable to 25 years vs 15 for LONGi
Shading	LONGi (narrow)	Publishes up to 70 % power recovery when shaded; AIKO gives no figure
Weight	AIKO Comet 2U	28.1 kg vs 28.5 kg

For a roof, the single-glass Guardian is the better buy: the dual-glass bifacial version adds 4 kg and gains little on a roof. Its 25 A max fuse also matches 25 A string fuses.

DC combiner set: fuse, breaker and SPD

Pick: Suntree. It is the only brand with all three parts in Thai stock and stated TUV/CE/CB marks. ABB has the best engineering margin but costs about 10 times more and is hard to buy in Thailand. Each set below covers one MPPT: 2 fuses with holders, 1 DC breaker and 1 DC SPD.

Brand	Fuse + holder	DC breaker	DC SPD Type 2, 1000 V	Notes
Suntree	SRD-30 holder + fuse	SL7N-63 2P 800 V (4P 1000 V available)	SUP2H1-PV 3P	IEC/EN 60947-2, TUV/CE/CB stated; good Thai stock
FEEO (budget)	FDS-32 + 25 A fuse	FPV-63 4P 1000 V	FSP-D40 3P 1000 V	No certificates cited in listings
CHINT	No genuine PV fuse found	NB1-63DC 4P 1000 V	No Thai listing	Grey-market in Thailand
Schneider	No Thai listing	C60PV-DC 2P 800 V	No Thai listing	Expensive
ABB	E90 PV holder	S802PV-S 2P 800 V	OVR PV T1-T2	Best margin, poor stock, about 10× the cost
TOMZN	Generic	TOB7Z-63 2P "1000 V"	TZG40-PV	2P 1000 V claim doubtful; avoid

Most 2-pole DC breakers from major brands are rated 800 V, not 1000 V. That is enough here: each 12-panel string reaches about 675 V on a cold 5 °C morning. Above 800 V, use a 4-pole 1000 V breaker.

AC combiner set: MCCBs, SPD and ATS

Pick: CHINT NXM-125S MCCBs (25 kA) with a Suntree SUP1-40 SPD and a Suntree SQ8P PC-class ATS. Schneider EasyPact EZC100H (30 kA) or ABB Formula give more margin but cost much more. A PC-class ATS (Suntree SQ8P, CSQ HYCQ4, Geya GATS) suits a grid/backup changeover better than a CB-class unit such as CHINT NXZ.

Combination	MCCB 4P 80 A / 50 A	AC SPD 4P 385 V	ATS 4P 100 A	Value
CHINT + Suntree	NXM-125S/4300, 25 kA	Suntree SUP1-40, 20/40 kA	Suntree SQ8P-100/4P, PC class	Best
CHINT + CSQ or Geya ATS	NXM-125S/4300, 25 kA	Suntree SUP1-40	CSQ HYCQ4-125/4 or Geya GATS-G-100, PC class	Good
CHINT only	NXM-125S/4300, 25 kA	NXU-IIG, 40 kA	NXZ-125/4, CB class	Fair (costly ATS)
ABB	Formula A1, 18–25 kA	OVR T2	—	Premium
Schneider	EasyPact EZC100H, 30 kA	iPRD40r	—	Premium, highest margin

Cables

Picks: Thai Yazaki THW for AC, and LINK PV1-F for DC. Choose Lapp or Phelps Dodge H1Z2Z2-K when a customer needs documented EN 50618 compliance.

Cable	Standard
Thai Yazaki THW 10 / 16 mm²	TIS 11 / IEC 60227
BCC THW 16 mm²	TIS certified
LINK PV1-F 4 mm²	Tinned Cu, XLPE, 1000/1500 V
Lapp Solar 4 mm²	EN 50618 / IEC 62930
Phelps Dodge H1Z2Z2-K	EN 50618

Mounting, conduit and grounding

Picks: Clenergy rails and clamps, Arrow IMC conduit (hot-dip galvanised inside and out; Panasonic as the UL-listed alternative), and Kumwell grounding. Clenergy is the only mounting brand on Thai retail shelves that states its alloy, wind standard and warranty. Kumwell is a Thai maker whose ground rod meets the 254 µm copper rule.

Item	Brand	Spec	Warranty	Best in
Rail and clamps	Clenergy PV-ezRack ECO	AL6005-T5, anodised ≥10 µm, AS/NZS 1170.2	12 yr	Overall, availability
Rail and clamps	Unbranded "TIT" kit	Alloy and wind rating not stated	—	Value
Rail and clamps	K2 Systems	Static calculation in K2 Base	12 yr, 20 yr if planned in K2 Base	Quality
Tile hook (ลอนคู่)	Eleckta / Daisentec / Lumira	Bracket for double-wave tile	—	—
IMC 1", 3 m	Arrow	Hot-dip galvanised inside and out; TIS 770	—	Best pick: rust protection, value
IMC 1", 3 m	Panasonic	In-line galvanised outside, epoxy-lined inside; TIS 770 + UL 1242	—	Documented UL listing, smooth pull
EMT 1", 3 m (indoor only)	Daiwa	Galvanised	—	—
Ground rod 5/8" × 10 ft	Kumwell GRCBU 5810	UL listed, ≥254 µm copper (UL 467)	—	Overall, value
Exothermic weld	Kumwell KW90 / KW115	—	—	Value
Exothermic weld	nVent Erico Cadweld	UL 467 benchmark	—	Quality
Inspection pit	Kumwell GXCIP concrete	310 × 310 × 192 mm	—	Durability

Use IMC outdoors and on the roof, and EMT only indoors.

Installation standards: IMC, EMT, conduit, enclosures, heat and terminals

Thai code rules marked [EIT] are from secondary sources describing EIT 2001-64 and the EIT rooftop PV standard (B.E. 2565); most match the NEC. Check clause numbers against a purchased copy before quoting them in a contract. **New:** TISI approved มอก. 62852 (PV DC connectors) and มอก. 61439 เล่ม 8 (PV DC assemblies, i.e. combiner boxes) as mandatory on 18 Aug 2026, expected in force from December 2026.

Item	Standard	Key requirement	On-site check
IMC	มอก. 770-2533, UL 1242, ANSI C80.6, NEC 342	Galvanised, threaded, ½"–4". Wall: ¾" 2.10 mm, 1" 2.35 mm. Indoor, outdoor, wet, embedded	มอก./UL print; caliper the wall; cold-galvanise cut threads; support every 3 m and within 0.9 m of boxes [EIT]; ≤360° of bends between pull points
EMT	มอก. 770, UL 797, ANSI C80.3, NEC 358	Thin wall 1.2–1.6 mm, not threadable. Thai practice: indoor/dry only; no burial, no slabs	Compression or set-screw fittings; ream cut ends; never exposed on roofs
RSC (heavy wall)	มอก. 770, UL 6, ANSI C80.1	All locations incl. burial; best for exposed roof and coastal runs	Hot-dip galvanising
Flexible metal / liquid-tight	UL 1 / UL 360	FMC dry and short drops only, not an earth path; LFMC for wet drops to inverter/battery	Clip every 1.5 m; UV-rated glands; respect bend radius
PVC / uPVC / HDPE	Yellow PVC sold to มอก. 216-2524 (status unconfirmed)	PVC degrades in UV: use UV-rated uPVC outdoors; HDPE for burial ≥0.6 m	Expansion fittings on hot roofs; supports every 0.9–1.5 m
Conduit fill	EIT table = NEC	1 cable 53 %, 2 cables 31 %, 3+ cables 40 %	Compute from cable outer diameter

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Item	Standard	Key requirement	On-site check
Combiner enclosure	IEC 62208, IEC 61439-1/-2/-8, IEC 60529, IEC 62262	IP65 outdoors (IP66 coastal/humid); IK08+ on roofs; DC rating ≥ cold Voc; rating plate	Bottom entry only; glands tight; blanks fitted; check plate
DC side in the box	IEC 62548-1:2023, IEC 60364-7-712	Double insulation; DC switch-disconnector to IEC 60947-3; fuses 1.5–2.4 × Isc; SPD above Voc; warning labels (cl. 10)	Load-break test; SPD window; labels on box and isolator
Heat: inverter	Solis S6 manuals	Single-phase: 250 mm clearance, 800 mm above floor, ≤40 °C ambient advised. S6-EH3P: 300 mm between units, 500 mm in front	Shade canopy if west sun; no attics or closed rooms; watch for temperature-derating events
Heat: battery	Solis IntelliHome manual	Operates −20 to 55 °C, best 10–45 °C, derates above 45 °C; **not outdoors within 500 m of the coast**	Check April ambient; out of sun; nothing stacked on it
Heat: breakers in boxes	IEC 60898 (30 °C ref.), IEC 60947	Derate at 45–55 °C box temperature per maker's table	Size the box for heat; shaded side; off hot roof sheet
Heat: assemblies	IEC 61439	Terminal rise ≤70 K at 35 °C ambient	Thermal scan at full sun: >10–15 K above similar terminals = investigate
PV connectors	IEC 62852 / มอก. 62852	Mated pair must be the same type and maker	No mixed brands; maker's crimp tool; pull test
Lugs and ferrules	IEC 61238-1-1	Tinned copper lug, matched die; ferrules on fine-stranded cable; no solder-tinned ends	Die number on crimp; lug size matches cable
Torque	Maker's values	Solis S6-EH3P: AC 4–5 N·m, PE 2 N·m	Calibrated driver; torque-mark each terminal

Most common mistakes in Thailand: EMT on roofs or in slabs; yellow PVC in the sun (cracks in 2–3 years); mixed MC4 brands and plier crimps (main cause of DC arcing); IP65 boxes with open entries or top entry; AC breakers on the DC side; fuses or SPD rated below cold Voc; inverters and batteries in sun or attics; no torque tool or ferrules; conduit over 40 % fill; missing PV warning labels.

Method: specifications are from the manufacturers' datasheets linked in each table. Star ratings and picks are Solar ESS's own assessment. For prices and a design for your site, book a consultation at www.solaress.org or LINE: solaress1.

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